TECHNICAL RESEARCH PAPER

Customer Segmentation & Churn Pattern Analytics in European Banking

A segmentation-driven analysis of 10,000 retail customers across France, Germany and Spain

10,000	20.4%	€185.6M
customers analysed	overall churn rate	balances held by churners

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Dataset	European Bank customer records (n = 10,000; snapshot year 2025)
Deliverables	Research paper · Streamlit dashboard · Executive summary

Abstract

Customer churn is one of the largest hidden costs in retail banking. This paper analyses 10,000 retail customers of a European bank in France, Germany and Spain to establish which customer groups are most likely to leave, how churn differs across geography, age and financial profile, and whether it is concentrated among high-value customers. Using segmentation-driven descriptive analytics supported by chi-square tests and effect-size measures, we find an overall churn rate of **20.4%** (2,037 customers holding €185.6 million in balances).

Churn is concentrated rather than uniform. Customers aged 46–60 churn at **51.1%** and make up **41.3%** of all churners while representing only 16.5% of the customer base. German customers churn at **32.4%**, roughly twice the rate in France (16.2%) and Spain (16.7%). Inactive members churn at 26.9% against 14.3% for active members, and customers holding three or more products churn at 85.9%. The combination of Germany, ages 46–60 and inactivity produces an **80.8%** churn rate, whereas credit score, tenure and credit-card ownership show little or no relationship with churn. High-value customers (balance ≥ €139,512) churn at 23.0%, and the 367 high-value churners account for 31.6% of all balances lost.

To move from segments to individual customers, a gradient-boosting model was trained to score each customer's churn risk. On held-out customers it achieves an AUC of **0.87**; the 10% of customers it ranks as riskiest churn at **85%**, and the top 20% contain **63%** of all churners. The findings are translated into prioritised retention recommendations and an interactive Streamlit dashboard that lets decision-makers explore churn by segment and score customers in real time.

Keywords: *customer churn, customer segmentation, retail banking, churn prediction, gradient boosting, retention analytics, Streamlit*

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1. Introduction

1.1 Background and context

Customer churn represents one of the largest hidden costs in retail banking. Every customer who leaves reduces the bank's lifetime value from that relationship, forces the bank to spend more on acquiring replacements, and makes revenue less predictable. Because acquiring a new customer is generally more expensive than keeping an existing one, even small improvements in retention can have a material financial effect.

Banks routinely track an aggregate churn rate, but an aggregate number cannot tell management where to act. It does not show which customer groups are most likely to leave, how churn differs across countries, age groups and financial profiles, or whether the customers being lost are the bank's most valuable ones. Without this granularity, churn-management strategies remain generic, reactive and inefficient.

1.2 Problem statement and objectives

Despite holding rich customer-level data, banks face three recurring challenges: identifying high-risk customer segments, understanding how churn differs by geography and demographics, and quantifying the financial profile of the customers who leave. This study addresses these gaps through systematic, segmentation-driven analytics.

Table 1. Project objectives

Primary objectives	Secondary objectives
Measure the overall churn rate	Understand churn among high-value customers
Identify churn distribution across customer segments	Evaluate engagement and tenure patterns
Compare churn behaviour across European regions	Support strategic planning and marketing decisions

1.3 Research questions

- **RQ1.** Which customer groups are most likely to churn?
- **RQ2.** How does churn differ across countries, age groups and financial profiles?
- **RQ3.** Is churn concentrated among high-value or low-value customers?
- **RQ4.** Which factors are most strongly associated with churn, and which are not, so that retention effort can be targeted?
- **RQ5.** Can an individual customer's churn risk be predicted accurately enough to prioritise retention outreach?

2. Data and Methodology

2.1 Dataset description

The analysis uses a customer-level dataset of **10,000 records** and 14 fields covering customers of a European retail bank in France, Germany and Spain (snapshot year 2025). Each row is one customer. The identifiers (CustomerId, Surname) are not used in the analysis. The target variable is **Exited**, which equals 1 when the customer has left the bank.

Table 2. Analytical fields

Field	Description
CreditScore	Customer credit score
Geography	Country of residence (France, Germany, Spain)
Gender	Male / Female
Age	Customer age in years
Tenure	Years with the bank (0–10)
Balance	Account balance (€)
NumOfProducts	Number of bank products held
HasCrCard	Credit-card ownership (1 = yes)
IsActiveMember	Activity indicator (1 = active)
EstimatedSalary	Estimated annual salary (€)
Exited	Churn indicator and target variable (1 = left the bank)

Data validation. Before analysis the dataset was checked for missing values (none found), duplicate rows (none found), valid categories for Geography and Gender, valid 0/1 coding for Exited and the binary indicators, and plausible numeric ranges for Age, Tenure, CreditScore and Balance. No records were removed.

2.2 Segmentation design

Customers were grouped along five dimensions so that churn could be compared on a like-for-like basis. Thresholds were chosen to give interpretable groups of meaningful size; they are analyst choices and can be adjusted in the dashboard.

Table 3. Segmentation scheme

Dimension	Segments	Definition
Geography	France, Germany, Spain	Country of residence
Age	<30, 30–45, 46–60, 60+	Age in years at snapshot
Credit score band	Low, Medium, High	Low < 580; Medium 580–669; High ≥ 670
Tenure group	New, Mid-term, Long-term	New 0–2 yrs; Mid-term 3–6 yrs; Long-term 7–10 yrs

Dimension	Segments	Definition
Balance segment	Zero, Low, High	Zero = €0; Low = below the median non-zero balance (€119,840); High = at or above it
High-value customer	Yes / No	Balance in the top 25% of non-zero balances (≥ €139,512); 1,596 customers

2.3 Analytical workflow and KPIs

The analysis followed five steps: (1) data ingestion and validation; (2) customer segmentation; (3) churn distribution analysis (overall rate, segment-wise rates, contribution of each segment to total churn, and churned-versus-retained profiles); (4) comparative demographic analysis (gender, geography–age interaction and financial stability versus churn); and (5) high-value customer churn analysis (high-balance churners, salary versus balance patterns, and balances at risk).

Four key performance indicators summarise the results and are calculated live in the dashboard:

Table 4. Key performance indicators

KPI	Definition	Result
Segment churn rate	Churned customers ÷ customers in the segment	20.4% overall
High-value churn ratio	Churn rate of high-value customers ÷ overall churn rate	1.13× (23.0% vs 20.4%)
Geographic risk index	Country churn rate ÷ overall churn rate (>1 = above-average risk)	Germany 1.59; Spain 0.82; France 0.79
Engagement drop indicator	Churn rate of inactive members ÷ churn rate of active members	1.88× (26.9% vs 14.3%)

2.4 Statistical testing

To distinguish genuine differences from random variation, associations between each categorical factor and churn were tested with Pearson's chi-square test of independence (Pearson, 1900). Because chi-square values grow with sample size, the strength of each association is reported with Cramér's V (Cramér, 1946), where values near 0.1 indicate a small association, near 0.3 a moderate one and above 0.5 a strong one (a common rule of thumb). Differences in continuous variables between churned and retained customers were tested with the Mann–Whitney U test (Mann & Whitney, 1947). Significance is judged at the 5% level. All results are descriptive associations and do not establish causation.

3. Findings

This section presents the results in the order of the analytical workflow. Each table reports the number of customers, the number who churned, the churn rate, and the segment's share of all churners (its "contribution" to total churn). Comparing the last column with the segment's share of customers shows where churn is over-represented.

3.1 Overall churn and customer profile

Of 10,000 customers, **2,037 (20.4%)** left the bank, roughly one in five. The departed customers held **€185.6 million** in account balances, 24.3% of all balances in the dataset. Churned customers are older on average (44.8 vs 37.4 years) and hold higher average balances (€91,109 vs €72,745) than retained customers; both differences are statistically significant (Mann–Whitney $p < 0.001$). Credit score differs only marginally (645 vs 652, $p = 0.020$), while estimated salary ($p = 0.227$) and tenure ($p = 0.162$) do not differ significantly.

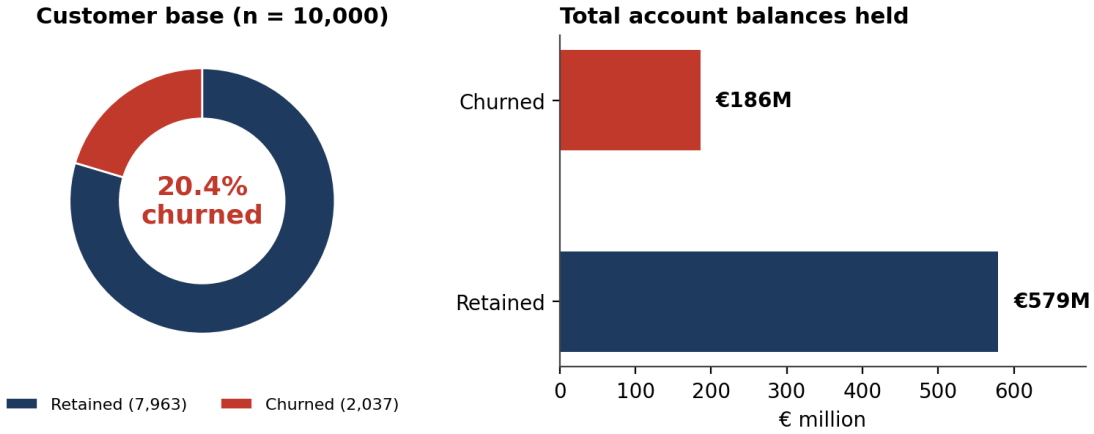


Figure 1. Overall churn rate and the balances held by churned versus retained customers.

Table 5. Average profile of retained versus churned customers

Measure	Retained	Churned	Significant difference?
Age (years)	37.4	44.8	Yes ($p < 0.001$)
Balance (€)	72,745	91,109	Yes ($p < 0.001$)
Credit score	651.9	645.4	Marginal ($p = 0.020$)
Estimated salary (€)	99,738	101,466	No ($p = 0.227$)
Tenure (years)	5.0	4.9	No ($p = 0.162$)

Source: author's analysis. Differences tested with the Mann–Whitney U test.

3.2 Geographic differences

Germany stands out. German customers churn at **32.4%**, almost double the rate in Spain (16.7%) and France (16.2%). Germany holds a quarter of the customer base but accounts for **40.0%** of churners, and its geographic risk index of 1.59 means it is 59% above the bank-wide average. In balance terms the gap is larger still: German churners took €98.0 million with them, 52.8% of all balances lost, against

€57.7 million for France and €29.9 million for Spain. The association between country and churn is statistically significant ($\chi^2 = 301.3$, $p < 0.001$, $V = 0.17$).

Table 6. Churn by country

Segment	Customers	Churned	Churn rate	Share of all churn
France	5,014	810	16.2%	39.8%
Germany	2,509	814	**32.4%**	40.0%
Spain	2,477	413	16.7%	20.3%

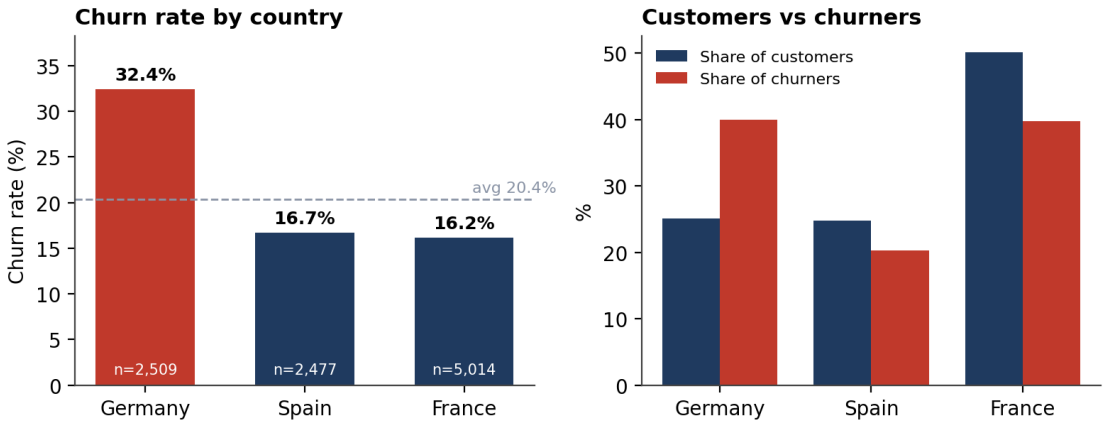


Figure 2. Churn rate by country (left) and each country's share of customers versus share of churners (right). Bars above the dashed line exceed the 20.4% average.

3.3 Age

Age is the single most informative demographic factor. Churn rises from 7.6% among customers under 30 and 15.3% among those aged 30–45 to **51.1%** for customers aged 46–60, then falls to 24.8% for those over 60. The 46–60 group is only 16.5% of customers yet supplies **41.3%** of all churners; the 30–45 group, although the largest, is under-represented among churners relative to its size. The association is strong ($\chi^2 = 1{,}231$, $p < 0.001$, $V = 0.35$).

Table 7. Churn by age group

Segment	Customers	Churned	Churn rate	Share of all churn
<30	1,641	124	7.6%	6.1%
30-45	6,248	956	15.3%	46.9%
46-60	1,647	842	**51.1%**	41.3%
60+	464	115	24.8%	5.6%

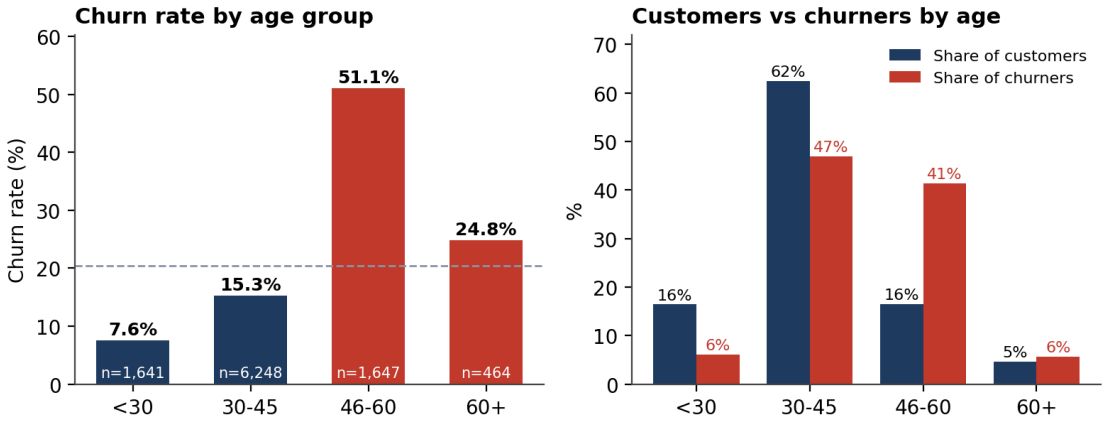


Figure 3. Churn rate by age group (left; dashed line = 20.4% average) and the age mix of the customer base versus the age mix of churners (right).

3.4 Geography–age interaction

Geography and age compound each other. Churn among customers aged 46–60 reaches **67.3% in Germany**, against 45.8% in France and 40.7% in Spain. German customers churn at higher rates in every age band; for example, German customers aged 30–45 (25.3%) churn at about double the rate of their French (11.9%) and Spanish (12.5%) peers. The German market therefore appears to have a country-wide problem that is most acute among middle-aged and older customers.

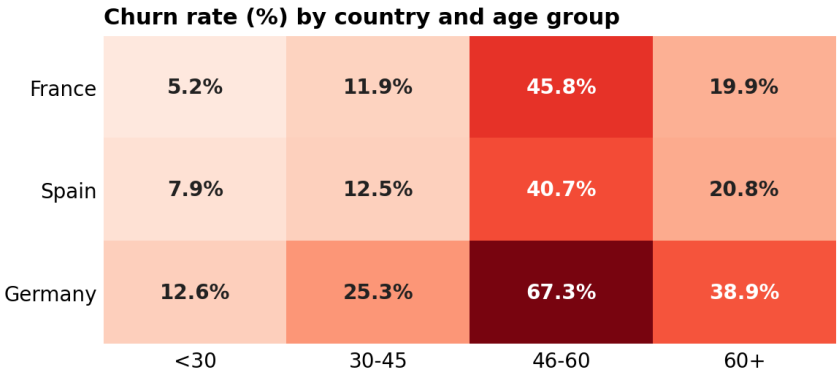


Figure 4. Churn rate (%) by country and age group. Darker cells indicate higher churn.

3.5 Gender

Female customers churn at **25.1%** against 16.5% for male customers ($\chi^2 = 112.9$, $p < 0.001$, $V = 0.11$), and women account for 55.9% of churners despite being 45.4% of customers. The gap is consistent across all three countries: France 20.3% vs 12.7%, Spain 21.2% vs 13.1% and Germany 37.6% vs 27.8% (female vs male). Because the gap appears everywhere, it is unlikely to be explained by the country mix alone.

Table 8. Churn by gender

Segment	Customers	Churned	Churn rate	Share of all churn
Female	4,543	1,139	25.1%	55.9%
Male	5,457	898	16.5%	44.1%

3.6 Engagement and product holding

Engagement. Inactive members churn at 26.9% versus 14.3% for active members, an engagement drop indicator of 1.88×. Inactive members are 48.5% of customers but 63.9% of churners, which makes inactivity the most actionable warning signal in the data because it can be monitored continuously.

Product holding. The number of products shows the strongest association with churn of any factor (V = 0.39). Customers with one product churn at 27.7%, but those with **two products churn at only 7.6%**, the lowest rate of any group in the study. Churn then rises sharply to 82.7% for three products and 100% for four. The three-and-four-product groups are small (326 customers in total, 85.9% churn), so the exact rates should be treated with caution, but the pattern is clear and warrants investigation.

Table 9. Churn by engagement and number of products

Segment	Customers	Churned	Churn rate	Share of all churn
Inactive	4,849	1,302	26.9%	63.9%
Active	5,151	735	14.3%	36.1%
1 product	5,084	1,409	27.7%	69.2%
2 products	4,590	348	**7.6%**	17.1%
3 products	266	220	82.7%	10.8%
4 products	60	60	100.0%	2.9%

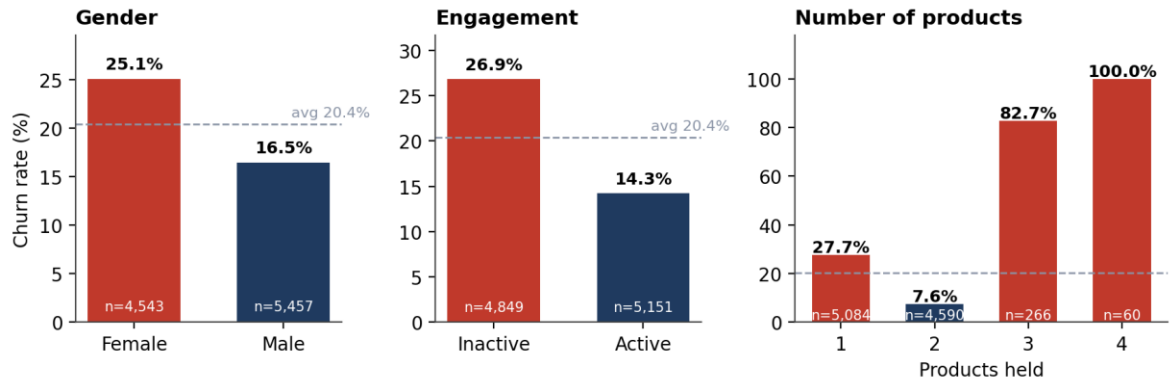


Figure 5. Churn rate by gender, engagement and number of products held (dashed line = 20.4% average).

> **Interpretation note**
> The product pattern is an association, not proof of cause. Possible explanations include bundling or pricing issues for multi-product customers, service problems, or customers consolidating accounts before leaving. The data cannot distinguish between these, so the bank should investigate them directly (see Section 6).

3.7 Credit score and tenure

Two factors often assumed to drive churn turn out to matter very little. Churn varies only from 19.3% (high credit score) to 22.0% (low credit score) across credit bands (V = 0.03), and from 19.5% (long-term) to 21.2% (new customers) across tenure groups, a difference that is not statistically significant

(p = 0.253). Credit-card ownership is equally uninformative (20.2% vs 20.8%, p = 0.492). These variables are therefore poor criteria for targeting retention effort.

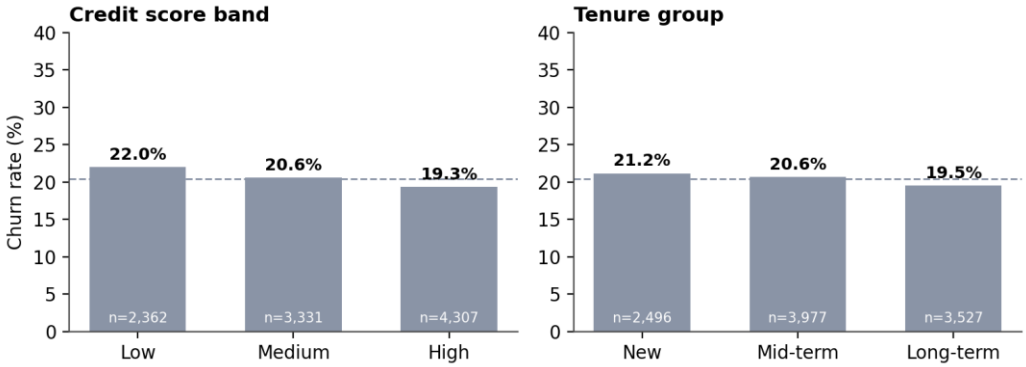


Figure 6. Churn rate by credit score band and tenure group. The dashed line marks the 20.4% average; all groups sit close to it.

3.8 Financial profile and high-value customers

Balance. Customers with a zero balance churn least (13.8%), while low- and high-balance customers churn at similar, higher rates (24.0% and 24.2%). This comparison must be read with care: **no German customer has a zero balance**, whereas 48% of French and Spanish customers do. Among customers with a positive balance, churn is 18.2% in France, 19.6% in Spain and 32.4% in Germany, so a substantial part of the balance effect is really the Germany effect.

Table 10. Churn by balance segment

Segment	Customers	Churned	Churn rate	Share of all churn
Zero-balance	3,617	500	13.8%	24.5%
Low-balance	3,191	765	24.0%	37.6%
High-balance	3,192	772	24.2%	37.9%

High-value customers. The 1,596 high-value customers (balance ≥ €139,512) churn at **23.0%** against 19.9% for all other customers, a high-value churn ratio of 1.13×. The 367 high-value customers who left took **€58.7 million**, which is 31.6% of all balances lost although they are only 18% of churners. Churn is therefore not confined to low-value customers; the bank is losing disproportionately valuable relationships.

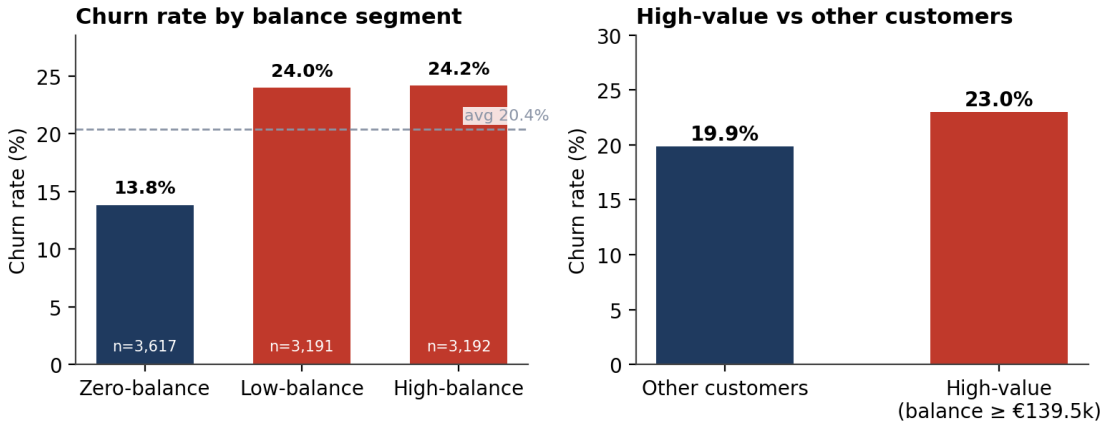


Figure 7. Churn rate by balance segment (left) and for high-value versus other customers (right).

Salary does not help to identify at-risk high-value customers: the correlation between estimated salary and balance is essentially zero (r = 0.013), and churned customers are spread across the whole salary range (Figure 8). Targeting should rely on behaviour (activity, products) and demographics rather than on salary.

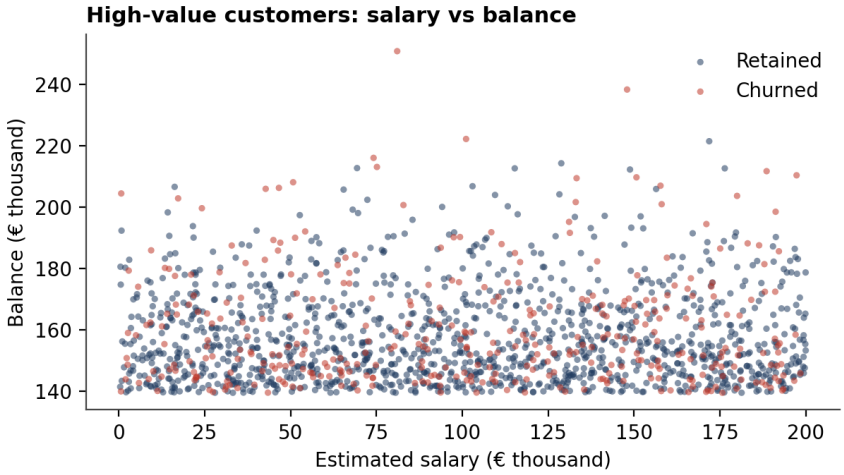


Figure 8. Salary versus balance for high-value customers. Churners (red) are spread across the salary range.

3.9 Combined high-risk segments

Combining factors isolates groups with extremely high churn. Among the segments examined (each with at least 250 customers), the highest-risk group is **German customers aged 46–60 who are inactive: 80.8% churn** (223 of 276). More broadly, inactive customers aged 46–60 churn at 68.4% regardless of country. At the opposite end, active French and Spanish customers under 30 churn at only 4.9% (649 customers). This spread, from about 5% to about 81%, shows how much the aggregate 20.4% hides.

Table 11. Highest-churn combined segments (minimum 250 customers)

Segment	Customers	Churned	Churn rate
Germany + Age 46-60 + Inactive	276	223	**80.8%**
Female + Age 46-60 + Inactive	414	299	72.2%
Germany + Female + Age 46-60	266	184	69.2%

Segment	Customers	Churned	Churn rate
Age 46-60 + Inactive	787	538	68.4%
Germany + Age 46-60	502	338	67.3%
Male + Age 46-60 + Inactive	373	239	64.1%
France + Age 46-60 + Inactive	345	210	60.9%
Female + Age 46-60	822	471	57.3%

3.10 Relative importance of factors

Figure 9 ranks individual segments by churn rate and Figure 10 ranks the factors by strength of association with churn. Number of products, age and geography have the strongest associations; active membership, balance segment and gender follow with small-to-moderate effects; credit band, tenure and credit-card ownership are close to irrelevant.

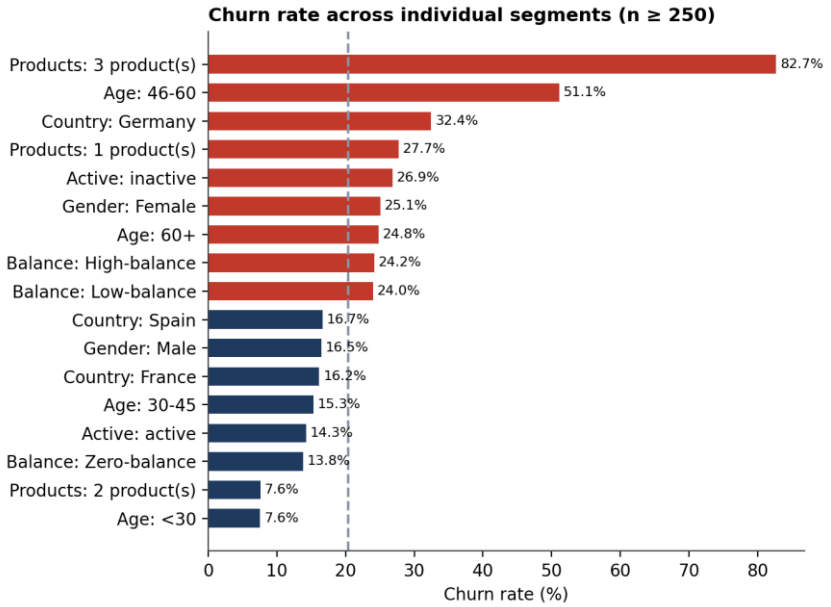


Figure 9. Churn rate for individual segments with at least 250 customers (red = above the 20.4% average).

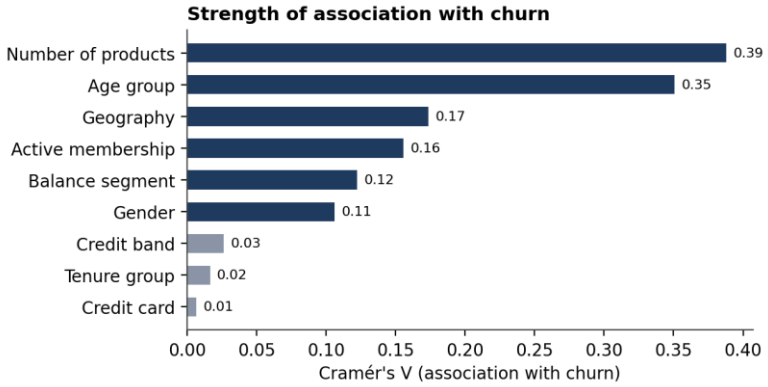


Figure 10. Strength of association between each factor and churn, measured by Cramér's V.

Table 12. Chi-square tests of association with churn

Factor	χ^2	df	p-value	Cramér's V
Number of products	1,503.6	3	<0.001	0.388
Age group	1,231.0	3	<0.001	0.351
Geography	301.3	2	<0.001	0.174
Active membership	243.0	1	<0.001	0.156
Balance segment	149.8	2	<0.001	0.122
Gender	112.9	1	<0.001	0.106
Credit score band	7.0	2	0.031	0.026
Tenure group	2.7	2	0.253	0.017
Has credit card	0.5	1	0.492	0.007

4. Predictive Modelling

Segment analysis shows where churn is concentrated, but retention teams act on individual customers. This section builds a customer-level churn score: a probability between 0 and 100% that each customer will leave, which can be used to rank customers and decide who to contact first.

4.1 Objective and approach

Two models were compared. **Logistic regression** is a transparent baseline that models churn as a weighted sum of customer characteristics. **Gradient boosting** builds many small decision trees in sequence, each correcting the errors of the previous ones (Friedman, 2001); it can capture non-linear patterns and interactions, such as risk that rises with age and then falls again, without the analyst specifying them in advance. Both were implemented in scikit-learn (Pedregosa et al., 2011).

The inputs were the ten customer attributes in Table 2. Identifiers (CustomerId, Surname) and the year field were excluded, and the outcome itself (Exited) is of course not an input. Number of products was treated as a category because its relationship with churn is not a straight line (Section 3.6). For the logistic model, a squared-age term and a zero-balance flag were added so that it can represent the age peak and the zero-balance group.

Table 13. Model specification

Item	Logistic regression	Gradient boosting
Algorithm	L2-regularised logistic regression (C = 1)	Histogram gradient boosting, 250 trees of depth 3
Learning rate / regularisation	–	0.06 / L2 = 1.0
Preprocessing	Numeric standardised; categorical one-hot encoded	Same pipeline
Extra features	Age² term, zero-balance flag	Same features (Age² and zero-balance flag included but not required)
Random seed	42	42

4.2 Validation design

To avoid overstating performance, 20% of customers (2,000, of whom 407 churned) were set aside as a **held-out test set** that was not used in any modelling decision; the models were trained on the remaining 8,000 customers. The split was stratified so that both parts have the same churn rate (20.4%). In addition, five-fold cross-validation on the training data gives an estimate of how much performance varies between samples.

Three measures are reported. **ROC AUC** is the probability that a randomly chosen churner is given a higher risk score than a randomly chosen non-churner (Hanley & McNeil, 1982); 0.5 is chance and 1.0 is perfect. **Average precision** summarises the trade-off between catching churners and raising false alarms; a model with no skill scores about 20.4%, the churn rate. The **Brier score** measures the

accuracy of the predicted probabilities themselves (lower is better; always predicting 20.4% scores 0.162).

4.3 Model performance

Table 14. Model performance

Model	Cross-validated AUC (training data)	Test AUC	Average precision	Brier score
Logistic regression	0.842 ± 0.012	0.845	0.671	0.108
Gradient boosting (selected)	0.866 ± 0.010	**0.868**	0.717	0.100

Test set: 2,000 customers not used for training. ± shows the standard deviation across five folds.

Both models discriminate well, and gradient boosting is clearly better: a test AUC of **0.868** versus 0.845, and an average precision of 0.717 against a no-skill baseline of 20.4%. The test result is almost identical to the cross-validated result (0.866), and the small fold-to-fold variation (± 0.010) indicates that performance is stable rather than the product of a lucky split. Gradient boosting was therefore selected. Its advantage over the simpler model comes from non-linear effects, notably the rise and fall of risk with age and the sharp differences between product counts.

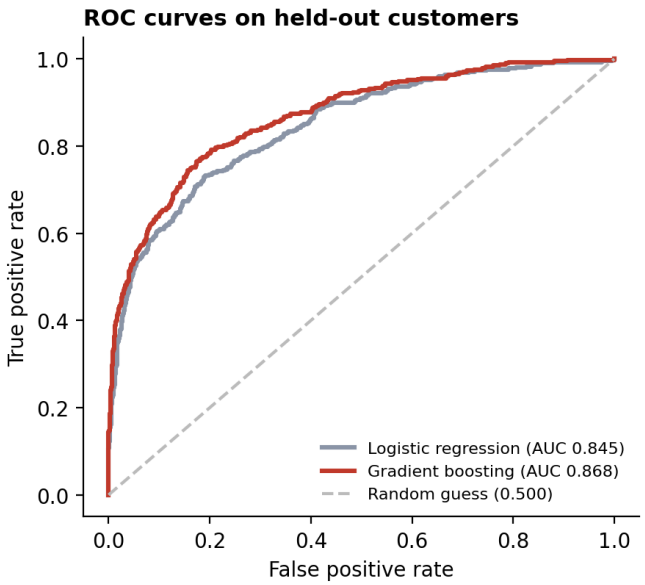


Figure 11. ROC curves for both models on the held-out test set. Curves closer to the top-left corner indicate better discrimination.

4.4 Business usefulness: gains and risk tiers

A model is useful if it lets the bank reach more churners with less effort. When test customers are ranked from highest to lowest predicted risk and split into ten equal groups (deciles), the riskiest decile has an actual churn rate of **85.0%**, which is 4.2 times the average (a lift of 4.2). Contacting only the top 20% of customers would reach **63.1%** of all churners, and the lowest three deciles contain customers who churn at just 3.0%. Predictions are therefore well ordered: higher scores correspond to higher actual churn, and the pattern holds on customers the model has never seen.

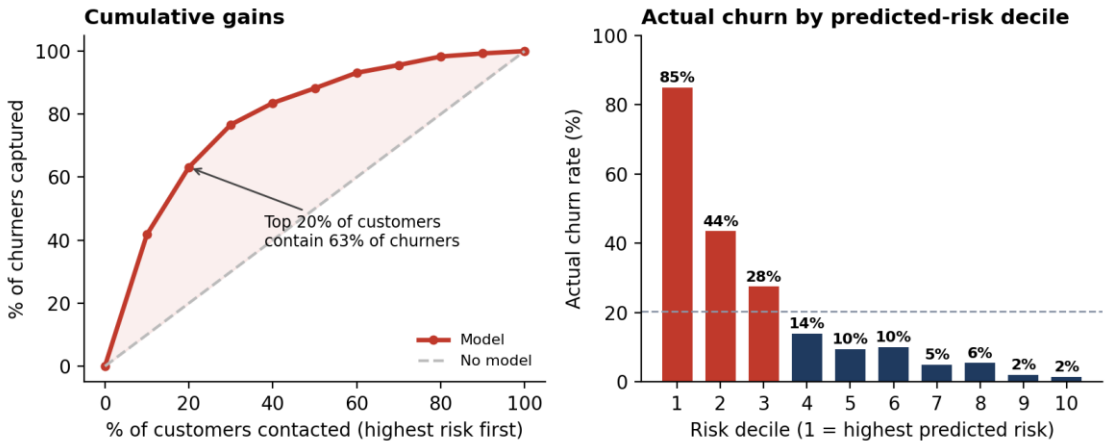


Figure 12. Cumulative gains (left) and actual churn rate by predicted-risk decile (right) on the test set. The dashed line marks the 20.4% average.

Table 15. Test-set results by predicted-risk decile

Decile	Customers	Churned	Churn rate	Cumulative % of churners captured	Lift
1	200	170	**85.0%**	41.8%	4.18×
2	200	87	**43.5%**	63.1%	2.14×
3	200	55	27.5%	76.7%	1.35×
4	200	28	14.0%	83.5%	0.69×
5	200	19	9.5%	88.2%	0.47×
6	200	20	10.0%	93.1%	0.49×
7	200	10	5.0%	95.6%	0.25×
8	200	11	5.5%	98.3%	0.27×
9	200	4	2.0%	99.3%	0.10×
10	200	3	1.5%	100.0%	0.07×

Decile 1 = customers with the highest predicted churn risk. Lift = decile churn rate ÷ overall churn rate.

For operational use, scores can be grouped into three **risk tiers**: High (predicted risk of 50% or more), Medium (20–50%) and Low (below 20%). The tiers are well separated on the test set:

Table 16. Risk tiers on the test set

Risk tier	Predicted risk	Customers	% of customers	Churned	Actual churn rate
High	≥ 50%	257	12.8%	199	**77.4%**
Medium	20–50%	384	19.2%	121	31.5%
Low	< 20%	1,359	68.0%	87	6.4%

The High tier is 12.8% of customers yet contains 199 of the 407 churners (49%), and those churners held €18.2 million, or 49.6% of all balances lost in the test sample. How aggressively to contact customers depends on the cost of an intervention relative to the value of a saved customer; Table 17 shows the trade-off at three cut-offs.

Table 17. Effect of the decision threshold on the test set

Contact customers with predicted risk ≥	Customers flagged	Precision (flagged who churn)	Recall (churners reached)	Churners caught	False alarms
30%	22.0%	60.3%	65.4%	266	175
40%	17.0%	68.7%	57.2%	233	106
50%	12.8%	77.4%	48.9%	199	58

Churners in the test set: 407. A lower threshold reaches more churners but contacts more customers who would have stayed.

4.5 What drives the predictions

The importance of each factor was measured by shuffling its values and observing how much the model's AUC falls; a large drop means the model relies on that factor. **Number of products** (AUC drop 0.121) and **age** (0.118) dominate, followed by **active membership** (0.031), **balance** (0.030) and **geography** (0.019); gender contributes a little (0.008). Credit score, tenure, credit-card ownership and salary contribute essentially nothing. This reproduces, with a completely different method, the ranking found in the segment analysis (Figure 10): the factors that matter are the same.

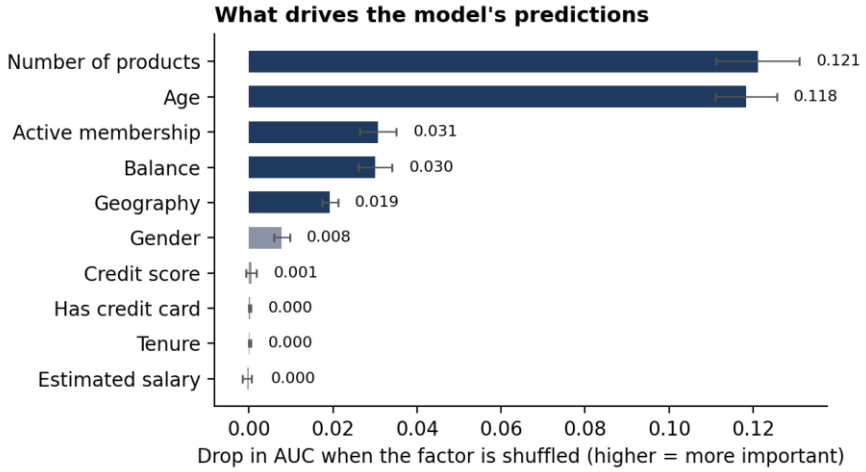


Figure 13. Permutation importance on the test set: reduction in AUC when each factor is shuffled (error bars show variation over 15 repetitions).

Figure 14 shows the average predicted risk if every customer were assigned a given value of one factor while everything else is left unchanged (a partial-dependence view; Friedman, 2001). Predicted risk is about 11% for customers in their early thirties, climbs steeply after age 40, peaks at about 55% around age 56, and then declines for the oldest customers. Predicted risk by product count (27%, 11%, 73% and 96% for one to four products) and by engagement (27% inactive versus 16% active) mirrors the raw results.

One result refines the geographic finding. Once the other characteristics are held constant, the German gap narrows from roughly double the French rate in the raw data (32.4% vs 16.2%) to about 1.4 times (24.6% vs 17.6%). Part of Germany's higher churn is therefore associated with other differences in its customer base, most visibly balances (no German customer has a zero balance), rather than with nationality alone. Germany nevertheless remains the highest-risk market after adjustment, so the recommendation to investigate it still stands.

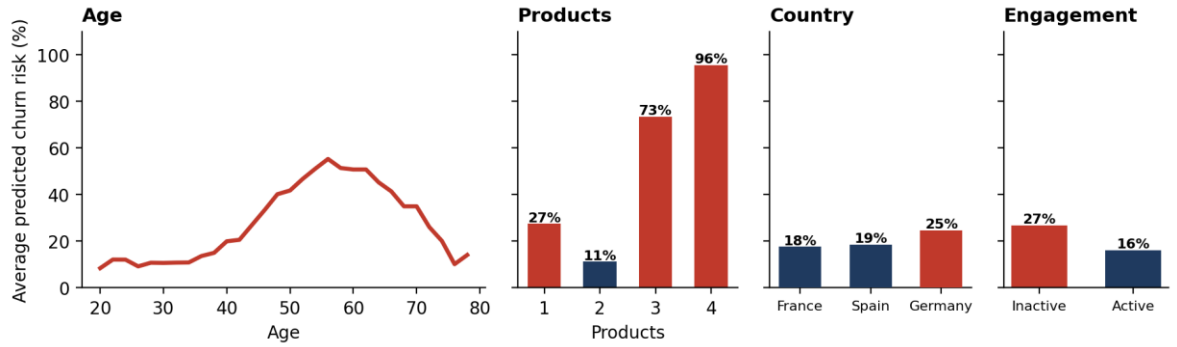


Figure 14. Average model-predicted churn risk when each factor is set to a given value for all customers (age, number of products, country and engagement).

4.6 Model limitations and responsible use

- **No test on future data.** The dataset has no dates, so the model could only be validated on a random hold-out sample from the same snapshot. Performance on future customers, after market or product changes, is unknown and should be monitored.
- **Prediction is not explanation.** The model ranks customers by risk but does not show why they will leave, and acting on a score does not guarantee that a customer is saved; interventions must be tested (Section 6).
- **Probabilities need monitoring.** The scores are reasonably accurate (Brier score 0.100 against 0.162 for a constant guess) but were not recalibrated; a production system should check calibration regularly.
- **Fairness and data protection.** The model uses personal attributes such as age and gender. Scores should be used to prioritise supportive outreach, not to restrict service or alter pricing, and the use of each attribute should be reviewed for fairness and data-protection compliance before deployment. Because gender adds only a little predictive power, the model could be retrained without it at a small cost in accuracy (test AUC 0.864 instead of 0.868).
- **Limited tuning.** Modest fixed settings were used rather than an extensive search, which keeps the estimates honest but means a small further gain may be possible.

4.7 Use in the dashboard

The dashboard includes a risk-scoring tab that trains the gradient-boosting model on the loaded dataset, assigns each customer a churn probability and risk tier, lists the highest-risk customers, and offers a what-if scorer for a single customer profile. It also shows the model's test-set AUC so that users can see how reliable the scores are.

5. Discussion

5.1 What the results mean

The central result is that churn in this bank is **concentrated and predictable from a few simple characteristics**. Three signals explain most of the exposure: where the customer lives (Germany), who the customer is (aged 46–60, and to a lesser extent female), and how the customer behaves (inactive, or holding one product versus two, or three or more). Characteristics that retention teams often use for targeting, such as credit score, tenure and card ownership, carry almost no information.

The balance findings change the commercial picture. Because churned customers hold higher balances on average and high-value customers churn more often than others, churn is not mainly a low-value problem. A retention strategy that focuses on protecting valuable relationships is likely to be justified on financial grounds alone.

The predictive model reinforces this picture in two ways. It ranks the same factors as most important, an independent confirmation from a different method, and it shows the pattern is strong enough to act on at customer level: contacting the riskiest fifth of customers reaches about 63% of likely leavers.

The German result is the single most important lead. A country-wide rate of 32.4%, elevated in every age and gender group, suggests market-level causes, such as competitive offers, pricing, product fit or service quality, rather than a customer-mix effect. The dataset cannot identify which, so a targeted investigation (for example customer interviews or exit surveys in Germany) is the logical next step.

5.2 Limitations

- **Single snapshot.** The data describe one point in time (2025) with no information on when customers left or on the observation window, so trends and seasonality cannot be assessed.
- **Association, not causation.** The analysis is descriptive. It identifies where churn is concentrated but not why customers leave; reasons such as pricing, complaints or competitor offers are not in the data.
- **Small groups.** The three- and four-product groups (326 customers) and the over-60 group (464 customers) are small, so their rates are less precise.
- **Analyst-defined thresholds.** Age, credit, tenure and balance cut-offs are reasonable but not unique; results for segment boundaries may shift if they are changed.
- **Confounding.** Some effects overlap, most visibly the balance effect and the absence of zero-balance customers in Germany. The multivariate model in Section 4 helps to separate overlapping effects.
- **Generalisability.** Results describe this customer base and should be validated on the bank's own production data before large-scale action.

5.3 Directions for future work

Three extensions would strengthen the work. First, **temporal validation**: if dated records become available, the model should be trained on earlier periods and tested on later ones to confirm it holds up over time. Second, **survival analysis** could model how long customers stay rather than whether

they leave. Finally, **controlled retention experiments** (A/B tests) are needed to measure whether the interventions proposed in Section 6 actually reduce churn, because a predictive score identifies who is at risk but not whether an intervention will work.

6. Recommendations

Recommendations are ordered by strength of evidence and potential financial benefit, each with a measurable success indicator. Suggested rollout: pilot rows 1–3 for 90 days with matched control groups, track the KPIs monthly in the dashboard, then extend to other segments, using the Section 4 churn score to rank customers within each.

Table 18. Prioritised retention recommendations

#	Target segment (evidence)	Recommended action	Success measure (suggested)
1	German customers aged 46–60 (502 customers; 67.3% churn; 80.8% if inactive)	Assign relationship managers for proactive outreach; review pricing, product fit and service in Germany; run exit interviews to find root causes.	Churn rate in this segment; German geographic risk index (target < 1.3)
2	Inactive members (4,849 customers; 26.9% churn; 63.9% of all churners)	Build an early-warning trigger on declining activity and launch re-engagement journeys (personal contact, relevant offers) before the customer leaves.	Share of inactive members; engagement drop indicator (target < 1.5×)
3	Customers with 3–4 products (326 customers; 85.9% churn)	Audit bundles, fees and service for multi-product customers and fix pain points; avoid pushing a third product until the cause is understood.	Churn rate of 3+ product customers
4	Single-product customers (5,084 customers; 27.7% churn)	Encourage a well-fitted second product (two-product customers churn at 7.6%); test through a controlled pilot to confirm the effect is causal.	Share moving to 2 products; churn of converted vs control group
5	High-value customers (1,596 customers; 23.0% churn; €58.7M lost)	Create a priority retention programme with dedicated contacts and early alerts on falling balances or activity.	High-value churn ratio (target ≤ 1.0×); balances retained
6	Female customers (25.1% vs 16.5% male; gap in all countries)	Investigate service, product and communication differences through customer research; adjust offers where gaps are found.	Female–male churn gap (percentage points)
7	Customers scored High risk by the model (predicted churn ≥ 50%; 13% of customers; 77% actually churned)	Score all customers monthly and contact the High tier first, then the Medium tier, within the segment programmes above; review the model quarterly.	Share of churners in contacted group; save rate versus a control group
8	Credit score, tenure, card ownership	De-prioritise as targeting criteria; they carry little information about churn risk.	Share of retention budget spent on high-signal segments

Illustrative impact (scenario, not a forecast)

Halving churn among the 787 inactive customers aged 46–60 (68.4% churn today) would retain about 269 customers; a 10-point reduction among German customers aged 46–60 would retain about 50. Actual results depend on how effective the interventions are and should be measured in a pilot first.

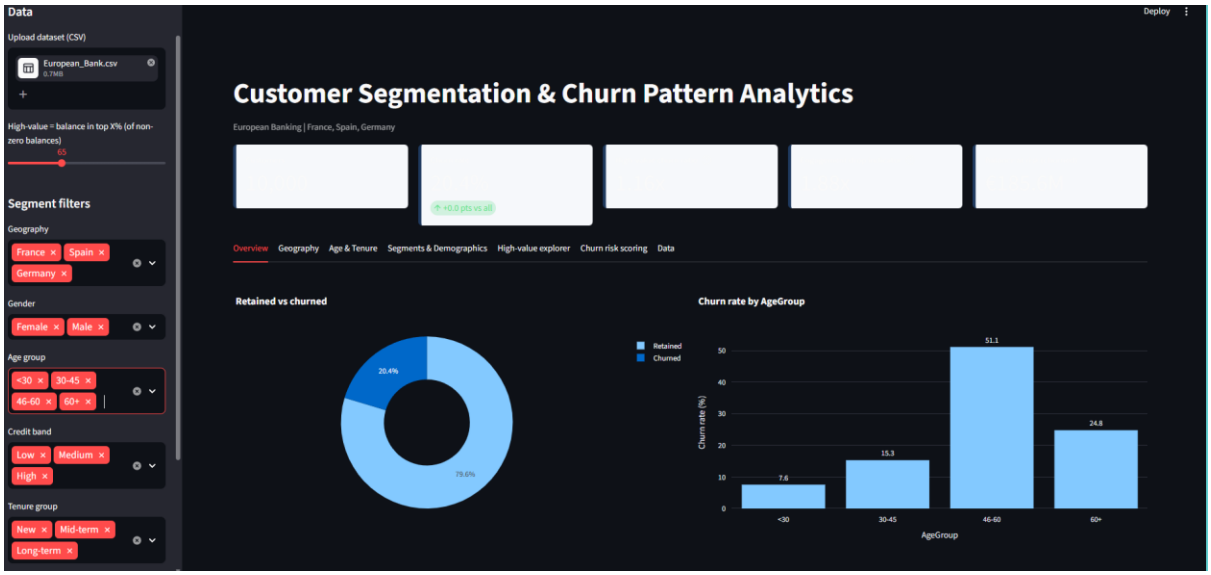
7. Interactive Dashboard

The findings are delivered together with a Streamlit web application so that decision-makers can explore churn themselves rather than relying on static tables. The dashboard reads the same dataset, applies the segmentation scheme in Table 3 and recalculates every KPI whenever a filter changes.

Table 19. Dashboard modules

Module	Purpose
KPI bar (always visible)	Customers, churn rate vs bank average, high-value churn ratio, engagement drop indicator, balances at risk; updates dynamically with filters
Overview	Retained vs churned split, segment-wise churn rates for any dimension, contribution of each segment to total churn, churned vs retained profiles
Geography	Churn by country, churners by country, geographic risk index, geography × age heat-map
Age & Tenure	Churn by age group, tenure group and years with the bank; age distributions of retained vs churned customers
Segments & Demographics	Gender, engagement, credit band and balance comparisons; balance distribution of churned vs retained customers
Churn risk scoring	Gradient-boosting churn probability for every customer, risk tiers (Low / Medium / High), a ranked list of the highest-risk customers, model performance and a what-if scorer for a single customer
High-value explorer	High-value churn rate, balances lost, salary-versus-balance view and a ranked list of high-value churners; the high-value threshold is adjustable
Data	Dataset quality summary and download of the filtered data

Users can filter by geography, gender, age group, credit band, tenure group and balance segment. The risk-scoring tab trains the model of Section 4 on the loaded dataset. Appendix B explains how to run the application.



8. Conclusion

This study set out to move beyond an aggregate churn rate and show where, among whom and at what financial cost customers leave. Of 10,000 customers, 20.4% churned, taking €185.6 million in balances. That average hides enormous variation: from about 5% for young, active French and Spanish customers to above 80% for inactive German customers aged 46–60.

Age, geography, product holding and engagement are the factors that matter most; credit score, tenure and card ownership matter very little. High-value customers churn more often than others and account for almost a third of the balances lost, so churn is a value problem as well as a volume problem.

The predictive model turns these insights into an operational tool: on held-out customers, a gradient-boosting score separates likely leavers from likely stayers well (AUC 0.87), so that contacting the riskiest fifth of customers reaches roughly 63% of those who would leave. These insights allow retention effort to be targeted at identifiable groups and individuals with measurable goals. The accompanying dashboard makes the analysis repeatable and accessible to non-technical stakeholders, and the next step is to validate the findings and the model through controlled pilots.

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Unified Mentor. Customer Segmentation & Churn Pattern Analytics in European Banking: Project documentation and requirements.

Appendix A: Full Segment Results

Churn results for every segment used in the analysis. Share of churn is the segment's proportion of all 2,037 churned customers; the overall churn rate is 20.4%.

Dimension	Segment	Customers	Churned	Churn rate	Share of churn
Country	France	5,014	810	16.2%	39.8%
	Germany	2,509	814	32.4%	40.0%
	Spain	2,477	413	16.7%	20.3%
Gender	Female	4,543	1,139	25.1%	55.9%
	Male	5,457	898	16.5%	44.1%
Age group	<30	1,641	124	7.6%	6.1%
	30-45	6,248	956	15.3%	46.9%
	46-60	1,647	842	51.1%	41.3%
	60+	464	115	24.8%	5.6%
Credit score band	Low	2,362	520	22.0%	25.5%
	Medium	3,331	685	20.6%	33.6%
	High	4,307	832	19.3%	40.8%
Tenure group	New	2,496	528	21.2%	25.9%
	Mid-term	3,977	821	20.6%	40.3%
	Long-term	3,527	688	19.5%	33.8%
Balance segment	Zero-balance	3,617	500	13.8%	24.5%
	Low-balance	3,191	765	24.0%	37.6%
	High-balance	3,192	772	24.2%	37.9%
Engagement	Inactive	4,849	1,302	26.9%	63.9%
	Active	5,151	735	14.3%	36.1%
Products	1	5,084	1,409	27.7%	69.2%
	2	4,590	348	7.6%	17.1%
	3	266	220	82.7%	10.8%
	4	60	60	100.0%	2.9%
Credit card	No card	2,945	613	20.8%	30.1%
	Has card	7,055	1,424	20.2%	69.9%

Appendix B: Running the Dashboard

The project folder contains app.py (the Streamlit application), utils.py (data loading, segmentation and KPI calculations), model.py (churn prediction models), requirements.txt and the dataset (Churn_Modelling.csv). To run the application:

- Install Python 3.9 or later.
- Open a terminal in the project folder and install the dependencies, then start the application with the two commands below.

```
pip install -r requirements.txt
streamlit run app.py
```

- The dashboard opens in the browser. If no dataset is found it shows a clearly labelled demo dataset; upload the real file in the sidebar or place it next to app.py.